

1.

$$f = \frac{\cos^2 \alpha H - \beta \sin^2 \alpha V}{\cos^2 \alpha + \beta \sin^2 \alpha},$$
$$X_0 \quad \text{normal coordinates} \quad \lambda \rightarrow \infty \quad t < 0 \quad \text{parabolic rescaling}$$

$$F_\lambda(x, t) = \lambda(F(x, T + \lambda^{-2}t) - X_0), \quad \Sigma_t^\lambda = F_\lambda(\Sigma, t).$$

$$H_\lambda = \lambda^{-1}H, \quad V_\lambda = \lambda^{-1}V, \quad f_\lambda = \lambda^{-1}f, \quad d\mu_t^\lambda = \lambda^2 d\mu_\tau,$$

$$\tau = T + \lambda^{-2}t$$

$$R > 0 \qquad -\infty < s_1 < s_2 < 0 \qquad \lambda \rightarrow \infty$$

$$\int_{s_1}^{s_2} \int_{\Sigma^\lambda \cap B_R(0)} |H_\lambda|^2 d\mu_t^\lambda dt \longrightarrow 0, \quad (5.1)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^\lambda \cap B_R(0)} |V_\lambda|^2 d\mu_t^\lambda dt \longrightarrow 0, \quad (5.2)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^\lambda \cap B_R(0)} |f_\lambda|^2 d\mu_t^\lambda dt \longrightarrow 0, \quad (5.3)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^\lambda \cap B_R(0)} |F_\lambda^\perp|^2 d\mu_t^\lambda dt \longrightarrow 0. \quad (5.4)$$

(5.5) Proposition 5.1 moving-cutoff displayed equation

2. cutoff weighted monotonicity

$$r > 0 \quad B_{2r}(X_0) \quad \text{normal-coordinate chart}$$

$$\phi \in C_c^\infty(B_{2r}(X_0)), \quad 0 \leq \phi \leq 1, \quad \phi \equiv 1 \quad \text{on } B_r(X_0).$$

$$\phi \qquad \qquad \qquad \lambda$$

$$\rho_{X_0,T}(F,\tau)=\frac{1}{4\pi(T-\tau)}\exp\left(-\frac{|F-X_0|^2}{4(T-\tau)}\right),$$

$$\text{Theorem 3.3} \qquad p \geq p_0$$

$$\begin{aligned}\Psi(\tau) &= \int_{\Sigma_\tau} (\cos \alpha)^{-p} \phi(F) \rho_{X_0,T}(F, \tau) \, d\mu_\tau, \\ Q(\tau) &= e^{c_1 \sqrt{T-\tau}} \Psi(\tau).\end{aligned}$$

Theorem 3.3

$$Q'(\tau) \leq -e^{c_1 \sqrt{T-\tau}} D(\tau) + c_2 e^{c_1 \sqrt{T-\tau}}, \tag{2.1}$$

$$D(\tau) \geq 0$$

$$\begin{aligned} D(\tau) &= \int_{\Sigma_\tau} (\cos \alpha)^{-p} \phi(F) \rho_{X_0,T}(F, \tau) \\ &\quad \cdot \left[\frac{1}{4} \left| \frac{(F - X_0)^\perp}{T - \tau} + f + H \right|^2 + |H + a(\alpha)V|^2 + \frac{7}{8}|H|^2 + \frac{7}{8}|V|^2 \right] d\mu_\tau. \end{aligned} \tag{2.2}$$

$$a(\alpha) \qquad \text{Theorem 3.3} \qquad \qquad \qquad (2.2) \qquad \qquad \qquad |H|^2, |V|^2$$

2.1

$$\tau_* < T$$

$$\widetilde{Q}(\tau)=Q(\tau)-c_2\int_{\tau_*}^{\tau}e^{c_1\sqrt{T-u}}\,du.$$

(2.1)

$$\widetilde{Q}'(\tau)\leq -e^{c_1\sqrt{T-\tau}}D(\tau)\leq 0.$$

$$Q(\tau) \geq 0 \qquad \qquad \qquad [\tau_*, T) \qquad \qquad \qquad \widetilde{Q} \qquad \qquad \qquad \widetilde{Q}(\tau) \quad \tau \uparrow T$$

$$L:=\lim_{\tau\uparrow T}Q(\tau)$$

$$(2.1) \qquad \text{one-sided inequality}$$

3. dissipation

$$\tau_i(\lambda) = T + \lambda^{-2}s_i, \quad i = 1, 2.$$

$$\lambda \quad 0 < \tau_1(\lambda) < \tau_2(\lambda) < T \quad T \quad (2.1) \quad \tau_1(\lambda) \quad \tau_2(\lambda)$$

$$\begin{aligned} \int_{\tau_1(\lambda)}^{\tau_2(\lambda)} e^{c_1\sqrt{T-\tau}} D(\tau) d\tau &\leq Q(\tau_1(\lambda)) - Q(\tau_2(\lambda)) \\ &+ c_2 \int_{\tau_1(\lambda)}^{\tau_2(\lambda)} e^{c_1\sqrt{T-\tau}} d\tau. \end{aligned} \quad (3.1)$$

$$L - L = 0$$

$$\tau_2(\lambda) - \tau_1(\lambda) = \lambda^{-2}(s_2 - s_1),$$

$$O(\lambda^{-2}) \quad \tau \quad T$$

$$\int_{\tau_1(\lambda)}^{\tau_2(\lambda)} D(\tau) d\tau \longrightarrow 0. \quad (3.2)$$

4. parabolic scaling

$$\tau = T + \lambda^{-2}t, \quad d\tau = \lambda^{-2}dt, \quad T - \tau = \lambda^{-2}(-t).$$

$$F - X_0 = \lambda^{-1}F_\lambda$$

$$\rho_{X_0,T}(F, T + \lambda^{-2}t) = \lambda^2 \rho_0(F_\lambda, t),$$

$$\rho_0(z, t) = \frac{1}{4\pi(-t)} \exp\left(-\frac{|z|^2}{4(-t)}\right). \quad (4.1)$$

$$\frac{(F - X_0)^\perp}{T - \tau} + f + H = \lambda \left(\frac{F_\lambda^\perp}{-t} + f_\lambda + H_\lambda \right). \quad (4.2)$$

$$(3.2) \quad d\mu_\tau = \lambda^{-2}d\mu_t^\lambda \quad d\tau = \lambda^{-2}dt \quad (4.1) \quad (4.2) \quad H = \lambda H_\lambda \quad V = \lambda V_\lambda \quad \lambda$$

$$\begin{aligned} &\int_{s_1}^{s_2} \int_{\Sigma_t^\lambda} (\cos \alpha_\lambda)^{-p} \phi(X_0 + \lambda^{-1}F_\lambda) \rho_0(F_\lambda, t) \\ &\cdot \left[\frac{1}{4} \left| \frac{F_\lambda^\perp}{-t} + f_\lambda + H_\lambda \right|^2 + |H_\lambda + a(\alpha_\lambda)V_\lambda|^2 + \frac{7}{8}|H_\lambda|^2 + \frac{7}{8}|V_\lambda|^2 \right] d\mu_t^\lambda dt \longrightarrow 0. \end{aligned} \quad (4.3)$$

estimate

5. scaled ball

$$R > 0 \quad \lambda > R/r \quad F_\lambda \in B_R(0)$$

$$X_0 + \lambda^{-1} F_\lambda \in B_{R/\lambda}(X_0) \subset B_r(X_0),$$

$$\phi(X_0 + \lambda^{-1} F_\lambda) = 1. \quad (5.1a)$$

$$t \in [s_1, s_2] \quad z \in B_R(0) \quad 0 < -s_2 \leq -t \leq -s_1 \quad (4.1)$$

$$\rho_0(z, t) \geq c_R := \frac{1}{4\pi(-s_1)} \exp\left(-\frac{R^2}{4(-s_2)}\right) > 0. \quad (5.1b)$$

$$\text{symplectic} \quad \text{Proposition 5.1} \quad \cos \alpha \geq \delta > 0 \quad (\cos \alpha_\lambda)^{-p} \geq 1$$

$$(4.3) \quad |H_\lambda|^2 \quad (5.1a) \quad (5.1b)$$

$$\frac{7}{8} c_R \int_{s_1}^{s_2} \int_{\Sigma_t^\lambda \cap B_R(0)} |H_\lambda|^2 d\mu_t^\lambda dt \leq (4.3) \quad \text{III}.$$

$$(5.1) \quad |V_\lambda|^2 \quad (5.2)$$

6. f_λ

$$f_\lambda = \frac{\cos^2 \alpha_\lambda H_\lambda - \beta \sin^2 \alpha_\lambda V_\lambda}{\cos^2 \alpha_\lambda + \beta \sin^2 \alpha_\lambda}.$$

$$\cos^2 \alpha_\lambda \geq \delta^2 \quad \beta \quad 0 \leq \sin^2 \alpha_\lambda, \cos^2 \alpha_\lambda \leq 1 \quad \beta, \delta$$

C_f

$$|f_\lambda|^2 \leq C_f (|H_\lambda|^2 + |V_\lambda|^2). \quad (6.1)$$

$$[s_1, s_2] \times (\Sigma_t^\lambda \cap B_R(0)) \quad (5.1) \quad (5.2) \quad (5.3)$$

7.

$$(4.3) \quad (5.1a) \quad (5.1b)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^\lambda \cap B_R(0)} \left| \frac{F_\lambda^\perp}{-t} + f_\lambda + H_\lambda \right|^2 d\mu_t^\lambda dt \longrightarrow 0. \quad (7.1)$$

$$t \in [s_1, s_2] \quad -t \leq -s_1$$

$$\begin{aligned} |F_\lambda^\perp|^2 &= (-t)^2 \left| \left(\frac{F_\lambda^\perp}{-t} + f_\lambda + H_\lambda \right) - f_\lambda - H_\lambda \right|^2 \\ &\leq 3(-s_1)^2 \left(\left| \frac{F_\lambda^\perp}{-t} + f_\lambda + H_\lambda \right|^2 + |f_\lambda|^2 + |H_\lambda|^2 \right). \end{aligned} \tag{7.2}$$

(7.2)

(7.1)

(5.3)

(5.1)

(5.4)

8.

equation (5.5)

(5.5)	$\phi_R(F_\lambda)$	scaled coordinates	cutoff	$\phi_R(\lambda(F - X_0))$	λ
		localized quantity			
		cutoff $\phi(F)$	scaled coordinates		
			$\phi(X_0 + \lambda^{-1}F_\lambda),$		
1	λ	$B_R(0)$	$B_R(0)$	weighted dissipation	
	L^2	(5.5)	moving-cutoff assertion	Proposition 5.1	

9.

Proposition 5.1					
1.	symplectic preservation	$\cos \alpha \geq \delta > 0$			
2.	Theorem 3.3	weighted monotonicity inequality			
3.	Proposition 5.1	parabolic scaling identities			
4.	elementary inequalities	Gaussian	cylinder		
	Lemma 5.2	Lemma 5.3	tangent-cone compactness	Allard theorem	
Proposition 5.1					
	Proposition 5.1	(5.1)-(5.4)	(5.5)	cutoff	
cutoff					